

Practice problems #10

1. Find all possible value(s) of h for which the vectors are linearly dependent:

$$\left\{ \begin{bmatrix} 1 \\ 3 \\ -3 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ h \end{bmatrix} \right\}.$$

2. True or False. Justify:

- (a) If $\{v_1, \dots, v_4\}$ is a set of linearly independent vectors in $\mathbf{R}^4$, then $\{v_1, v_2, v_3\}$ is also linearly independent.
- (b) The set of 5 vectors in $\mathbf{R}^4$ is linearly dependent.
- (c) The set of 4 vectors in $\mathbf{R}^5$ is linearly independent.

3. Verify geometrically that the given T is a linear transformation:

$T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ is a reflection about the line $x_2 = x_1$.

Answers are on the back.

#1: $h = -7$

#2(a): T (Why?)

#2(b): T (Why?)

#2(c): F (Why?)

#3: Sketch as we did in class.